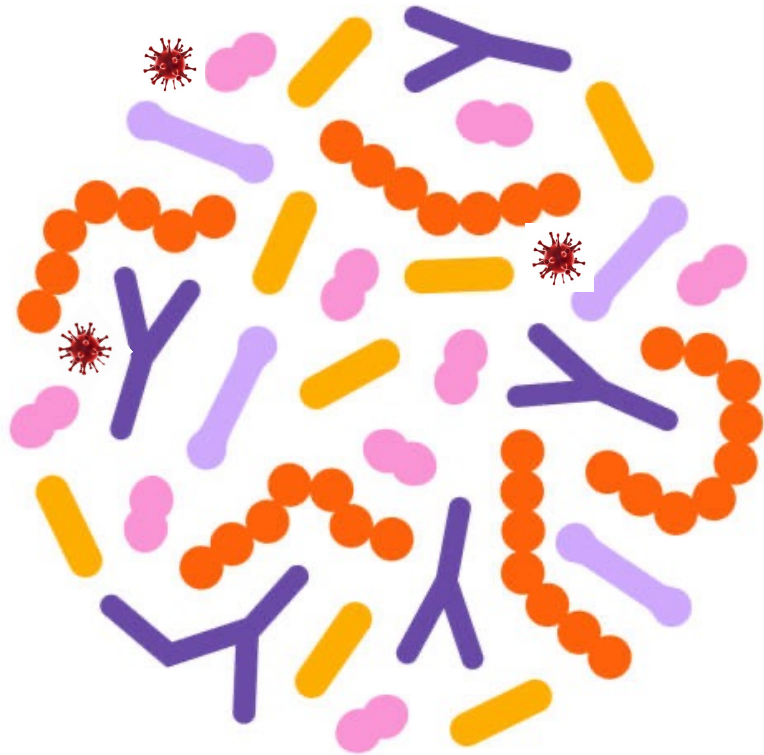




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# Introduction to Metagenomics

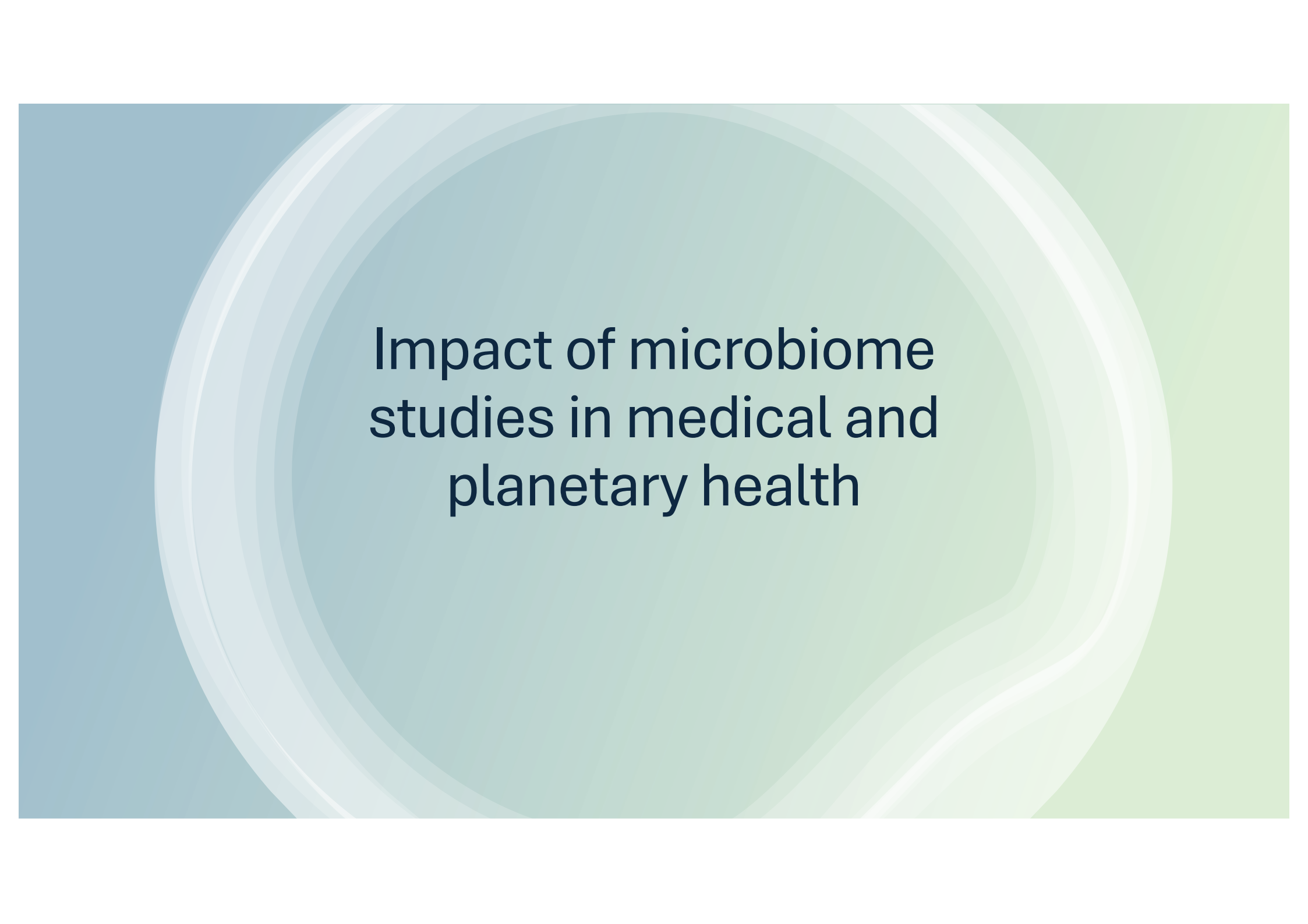
Data Science platform

DTU – Biosustain

Spring Data-type Courses

11th June 2025

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# Impact of microbiome studies in medical and planetary health

## Role of the intestinal microbiome

### Disease modifier

#### Cardiometabolic disorders

- Obesity
- Insulin resistance
- Type 2 diabetes mellitus
- NAFLD/NASH
- Atherosclerosis
- Hypertension
- Idiopathic thrombocytopenic purpura

#### Gastrointestinal disorders

- Clostridium difficile infection
- Ulcerative colitis
- Crohn's disease
- Chronic pouchitis
- Irritable bowel disease
- Idiopathic constipation
- Antibiotic resistant pathogens

#### Neuropsychiatric disorders

- Encephalopathy
- Autism spectrum disorders
- Multiple sclerosis
- Parkinson's disease
- Chronic fatigue syndrome
- Neurodegenerative disorders
- Appetite disorders
- Depression & anxiety

#### Immunologic disorders

- Graft-versus-host disease
- Multiple organ dysfunction syndrome
- Colonic cancer
- Arthritis
- Allergy
- Eczema
- Celiac disease

# Microbiome has been associated with a variety of diseases

- For most of the diseases it is not known whether the microbiota is the cause or the consequence of the disease
- For some diseases it may also be a disease modifier or may metabolize the drugs administered to the diseased individuals

Causal relationship

# Obesity associated gut microbiome – a transmissible trait

## An obesity-associated gut microbiome with increased capacity for energy harvest

Peter J. Turnbaugh, Ruth E. Ley, Michael A. Mahowald, Vincent Magrini, Elaine R. Mardis & Jeffrey I. Gordon 

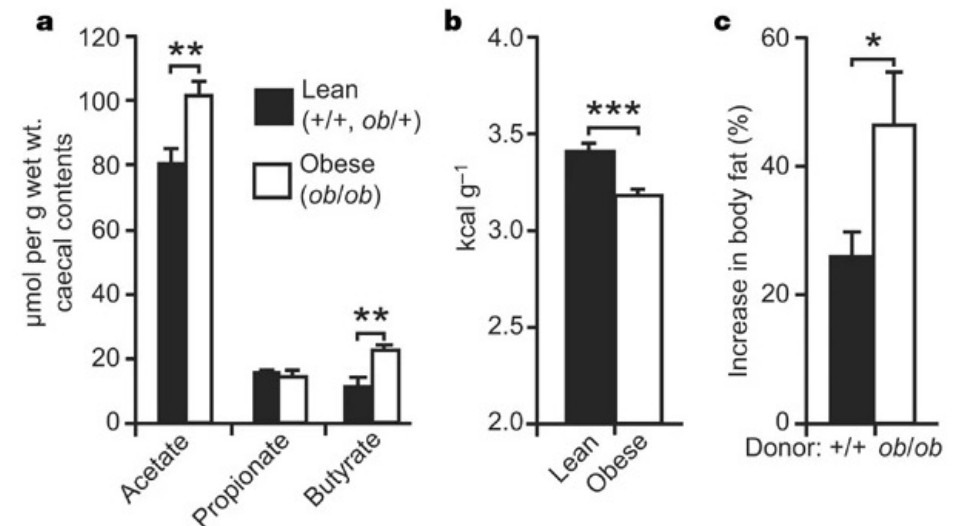
*Nature* 444, 1027–1031 (2006) | [Cite this article](#)

135k Accesses | 9059 Citations | 1196 Altmetric | [Metrics](#)

<https://doi.org/10.1038/nature05414>

- Obese microbiome has an increased capacity to harvest energy from the diet
- **The trait is transmissible!**  
Colonization of germ-free mice with an ‘obese microbiota’ results in a significantly greater increase in total body fat than colonization with a ‘lean microbiota’

**Figure 3: Biochemical analysis and microbiota transplantation experiments confirm that the *ob/ob* microbiome has an increased capacity for dietary energy harvest.**



Genetically obese mice had an increased concentration of the major fermentation end-products: butyrate and acetate

They also had significantly less energy remaining in their faeces

Germ-free mice colonized with an obese mice microbiota exhibited significantly greater increase in body fat

# The Human Microbiome Project (HMP)

Article | [Open access](#) | Published: 13 June 2012

## Structure, function and diversity of the healthy human microbiome

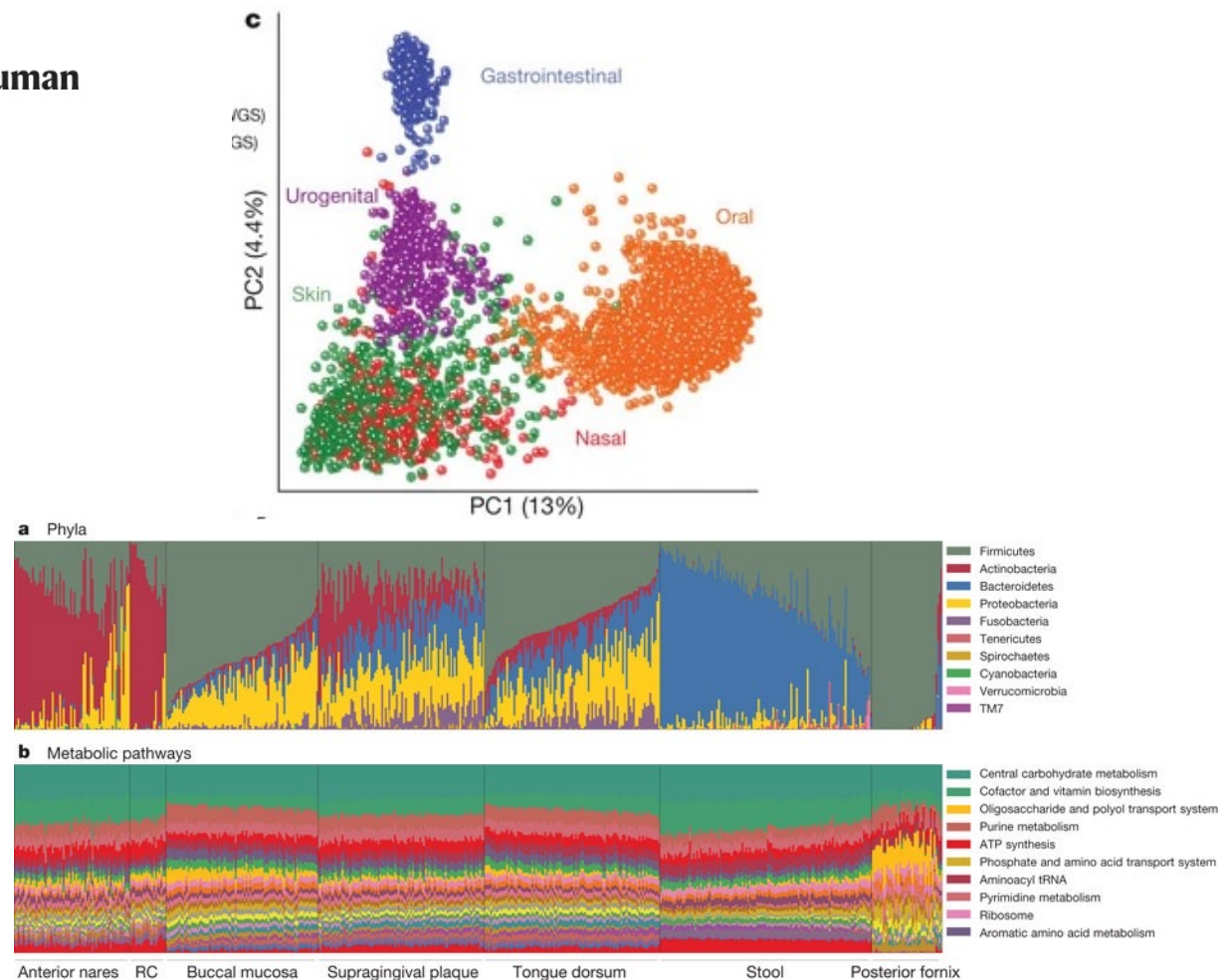
[The Human Microbiome Project Consortium](#)

[Nature](#) 486, 207–214 (2012) | [Cite this article](#)

450k Accesses | 1240 Altmetric | [Metrics](#)

<https://doi.org/10.1038/nature11234>

- Great initial resource to study the microbiome.
- Diversity of the human microbiome is concordant among measures, unique to each individual, and strongly determined by microbial habitat
- Carriage of microbial taxa varies while metabolic pathways remain stable within a healthy population



# Richness of human gut microbiome correlates with metabolic markers (lessons of MetaHIT study)

Article | Published: 28 August 2013

## Richness of human gut microbiome correlates with metabolic markers

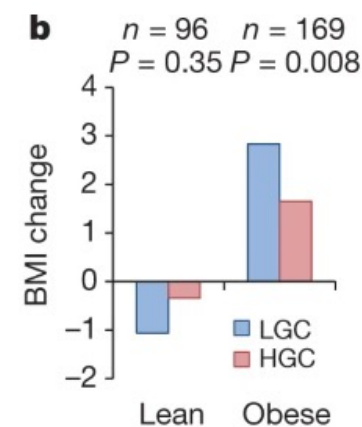
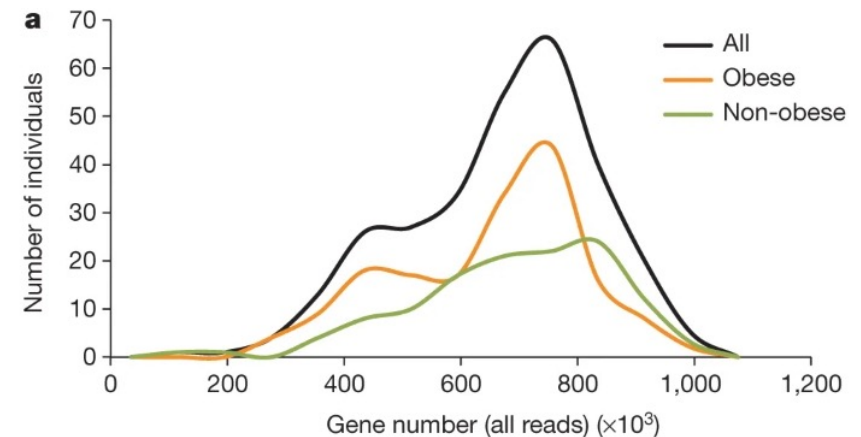
[Emmanuelle Le Chatelier](#), [Trine Nielsen](#), [Junjie Qin](#), [Edi Prifti](#), [Falk Hildebrand](#), [Gwen Falony](#), [Mathieu Almeida](#), [Manimozhiyan Arumugam](#), [Jean-Michel Batto](#), [Sean Kennedy](#), [Pierre Leonard](#), [Junhua Li](#), [Kristoffer Burgdorf](#), [Niels Grarup](#), [Torben Jørgensen](#), [Ivan Brandslund](#), [Henrik Bjørn Nielsen](#), [Agnieszka S. Juncker](#), [Marcelo Bertalan](#), [Florence Levenez](#), [Nicolas Pons](#), [Simon Rasmussen](#), [Shinichi Sunagawa](#), [Julien Tap](#), [MetaHIT consortium](#), ... [Oluf Pedersen](#)  [+ Show authors](#)

[Nature](#) 500, 541–546 (2013) | [Cite this article](#)

[10.1038/nature12506](https://doi.org/10.1038/nature12506)

- Individuals with a low bacterial richness are characterized by more marked overall adiposity, insulin resistance and dyslipidaemia and a more pronounced inflammatory phenotype
- Individuals with lower number of genes (gene richness) and higher BMI had less efficient weight loss when following a diet

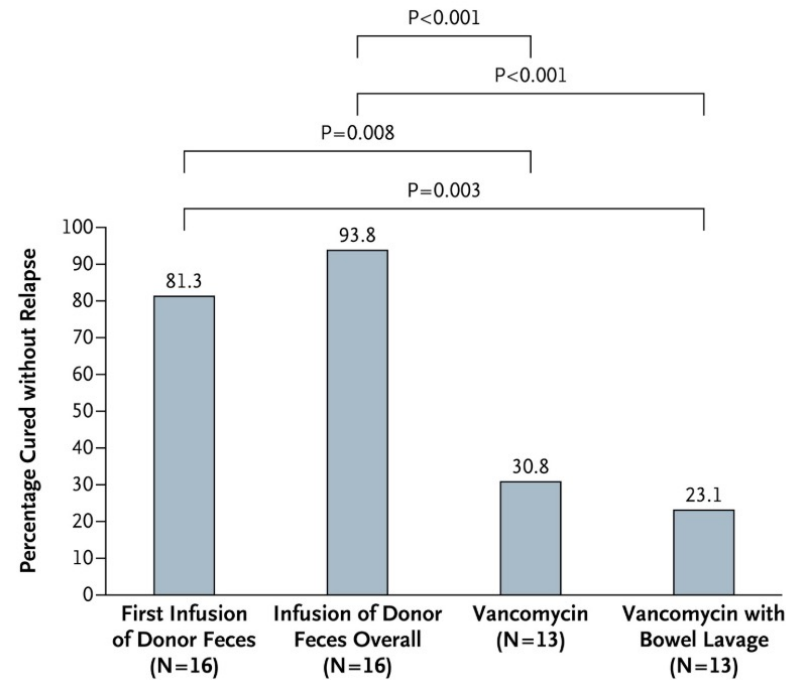
**Figure 1: Distribution of low and high gene count individuals ( $n = 292$ ).**



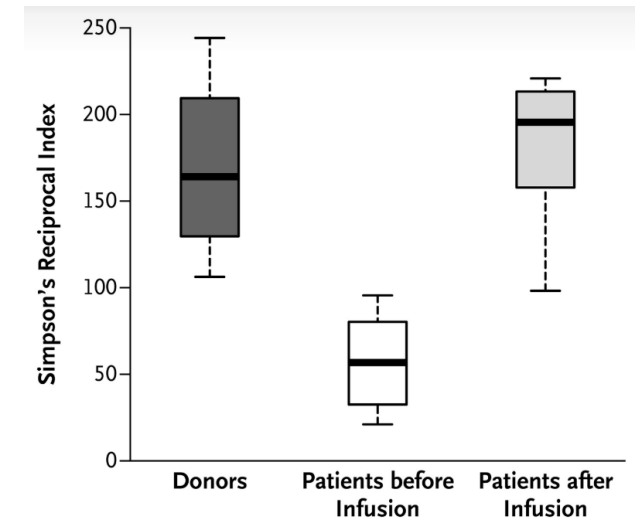
# Fecal material transplantation (FMT) as experimental cure for *Clostridioides difficile*



[10.1056/NEJMoa1205037](https://doi.org/10.1056/NEJMoa1205037)

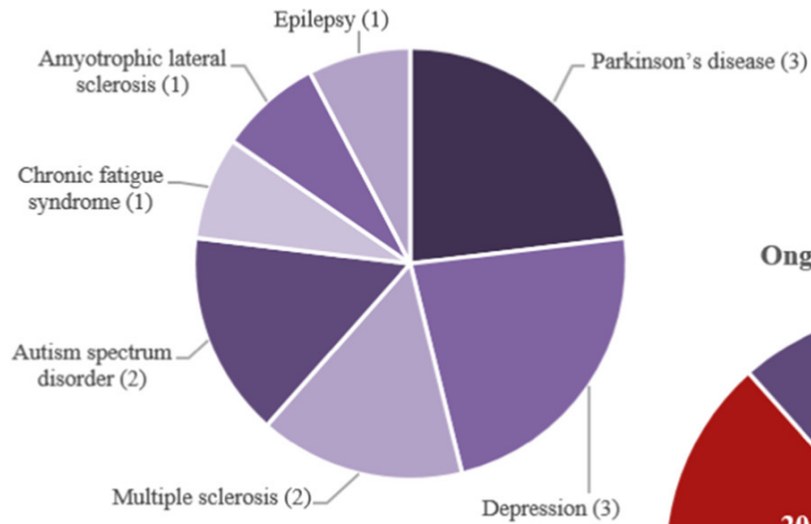


FMT successfully used to cure *Clostridioides difficile* infection with higher percentages of cured patients without relapse compared to the standard treatment (vancomycin)

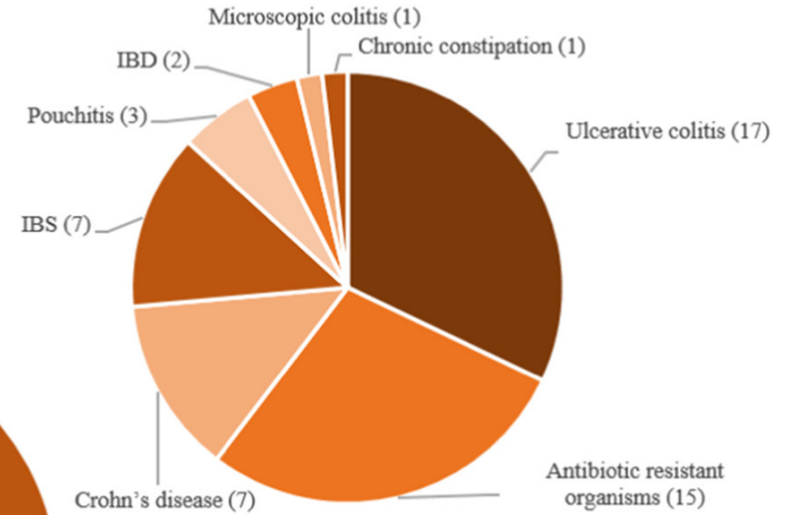


Diversity (Simpson's index) of patients after FMT resemble more the donors diversity

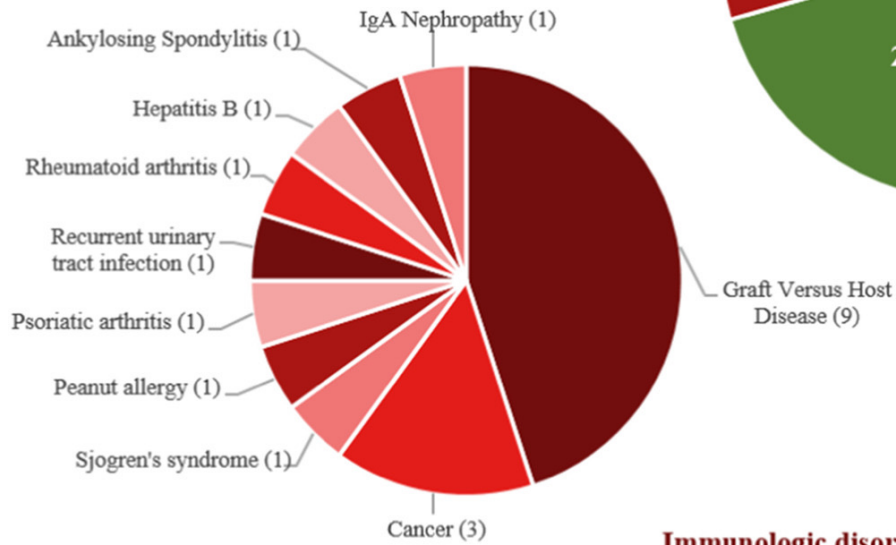
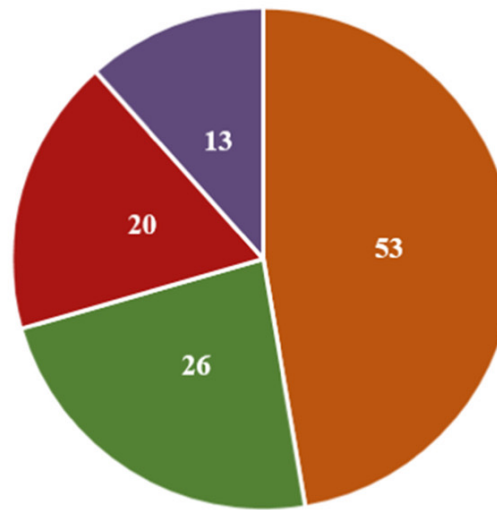
### Neuropsychiatric disorders



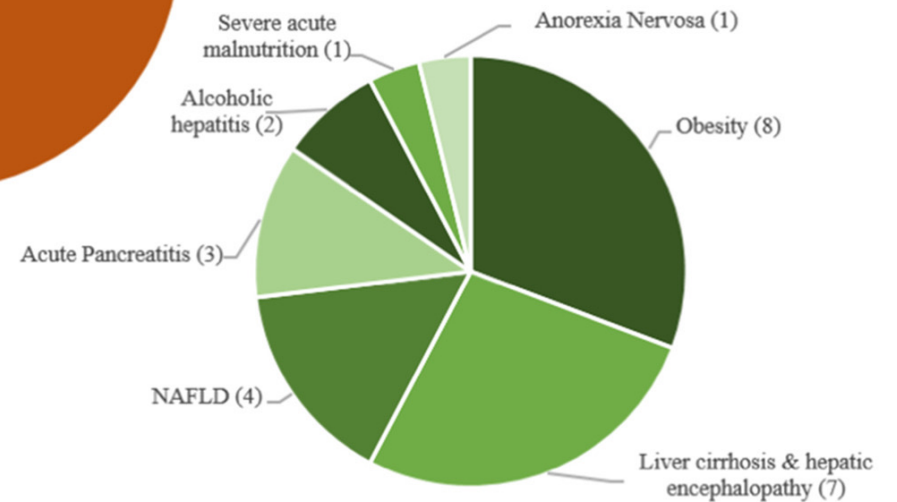
### Gastrointestinal disorders



### Ongoing trials with FMT



### Immunologic disorder



### Metabolic disorders

# Resident gut bacteria can affect patient responses to cancer immunotherapy

REPORT



## Gut microbiome influences efficacy of PD-1–based immunotherapy against epithelial tumors

BERTRAND ROUTY , EMMANUELLE LE CHATELIER , LISA DEROSA , CONNIE P. M. DUONG , MARYAM TIDJANI ALOU , ROMAIN DAILLÈRE , AURÉLIE FLUCKIGER , MERIEM MESSAOUDENE , CONRAD RAUBER, [...] AND LAURENCE ZITVOGEL +38 authors [Authors Info & Affiliations](#)

SCIENCE • 2 Nov 2017 • Vol 359, Issue 6371 • pp. 91–97 • DOI: 10.1126/science.aan3706 [DOI: 10.1126/science.aan3706](https://doi.org/10.1126/science.aan3706)

REPORT



## The commensal microbiome is associated with anti–PD-1 efficacy in metastatic melanoma patients

VYARA MATSON , JESSICA FESSLER , RIYUE BAO , TARA CHONGSUWAT , YUANYUAN ZHA , MARIA-LUISA ALEGRE , JASON J. LUKE , AND THOMAS F. GAJEWSKI [Authors Info & Affiliations](#)

SCIENCE • 5 Jan 2018 • Vol 359, Issue 6371 • pp. 104–108 • DOI: 10.1126/science.aao3290 [DOI: 10.1126/science.aao3290](https://doi.org/10.1126/science.aao3290)

REPORT



## Gut microbiome modulates response to anti–PD-1 immunotherapy in melanoma patients

V. GOPALAKRISHNAN , C. N. SPENCER , L. NEZI , A. REUBEN , M. C. ANDREWS , T. V. KARPINETS , P. A. PRIETO, D. VICENTE , K. HOFFMAN , [...] AND J. A. WARGO +60 authors [Authors Info & Affiliations](#)

SCIENCE • 2 Nov 2017 • Vol 359, Issue 6371 • pp. 97–103 • DOI: 10.1126/science.aan4236 [DOI: 10.1126/science.aan4236](https://doi.org/10.1126/science.aan4236)

In a series of back-to-back papers in Science:

- Routy et al. showed that abx consumption --> poor response to cancer immunotherapy (immunotherapeutic PD1 blockade)
- Low levels of *Akkermansia muciniphila* in lung and kidney cancer patients → oral supplementation of this bacteria to abx-treated mice restored response to immunotherapy
- Matson et al. and Gopalakrishnan et al. studied melanoma patients receiving PD-1 blockade and found a greater abundance of “good” bacteria in the guts of responding patients.
- Non-responders had an imbalance in gut flora composition, which correlated with impaired immune cell activity.

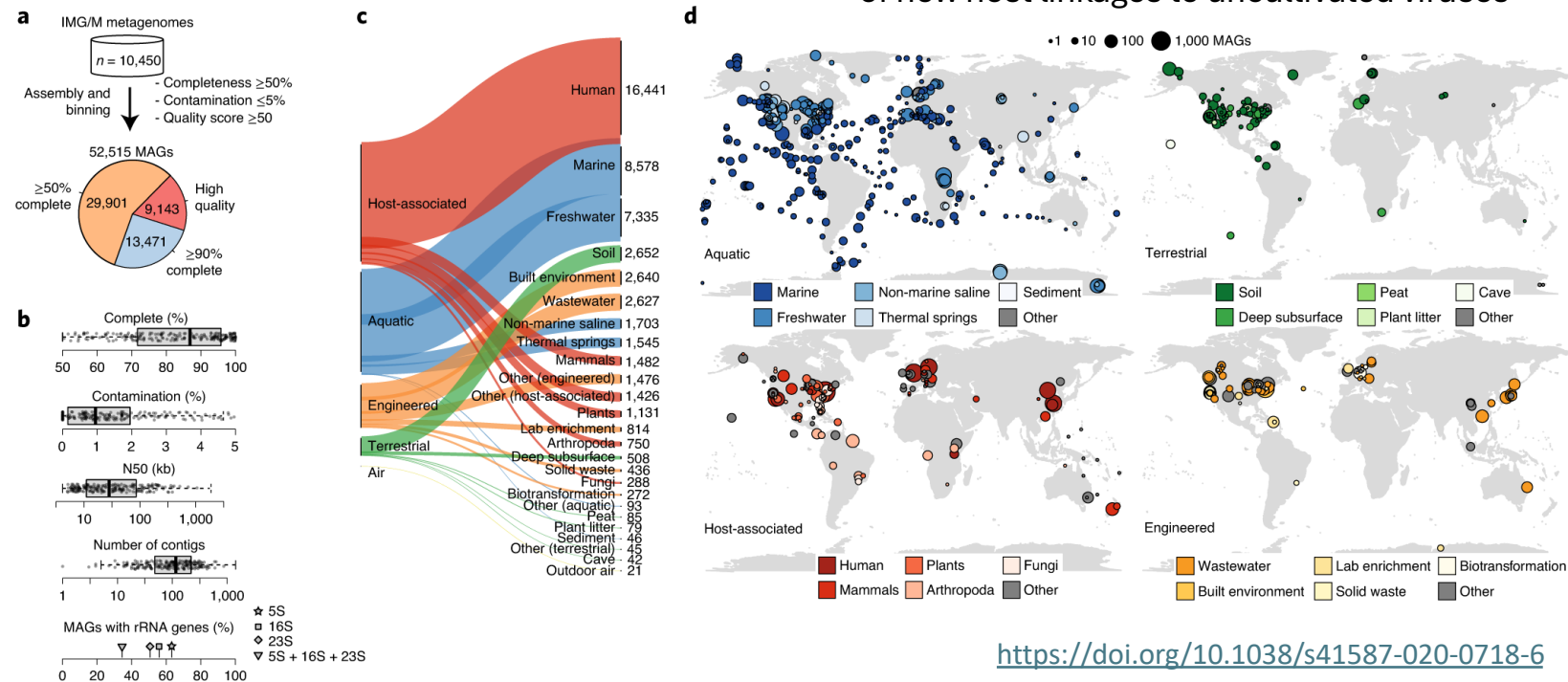
# A genomic catalog of Earth's microbiome

## A genomic catalog of Earth's microbiomes

Stephen Nayfach, Simon Roux, Rekha Seshadri, Daniel Udwarý, Neha Varghese, Frederik Schulz, Dongying Wu, David Paez-Espino, I-Min Chen, Marcel Huntemann, Krishna Palaniappan, Joshua Ladau, Supratim Mukherjee, T. B. K. Reddy, Torben Nielsen, Edward Kirton, José P. Faria, Janaka N. Edirisinghe, Christopher S. Henry, Sean P. Jungbluth, Dylan Chivian, Paramvir Dehal, Elisha M. Wood-Charlson, Adam P. Arkin, IMG/M Data Consortium, ... Emile A. Elie-Fadrosch ✉ + Show authors

*Nature Biotechnology* 39, 499–509 (2021) | [Cite this article](#)

- >10,000 metagenomes collected from diverse habitats comprising host associated, aquatic, terrestrials, air and engineered microbiomes
- 52,515 metagenome-assembled genomes representing 12,556 novel candidate species from 135 different phyla
- Performing metabolic modelling, understanding secondary-metabolite biosynthetic potential and for resolving thousands of new host linkages to uncultivated viruses



<https://doi.org/10.1038/s41587-020-0718-6>

# Antimicrobial resistance is a major threat to global health!

## Genomic analysis of sewage from 101 countries reveals global landscape of antimicrobial resistance

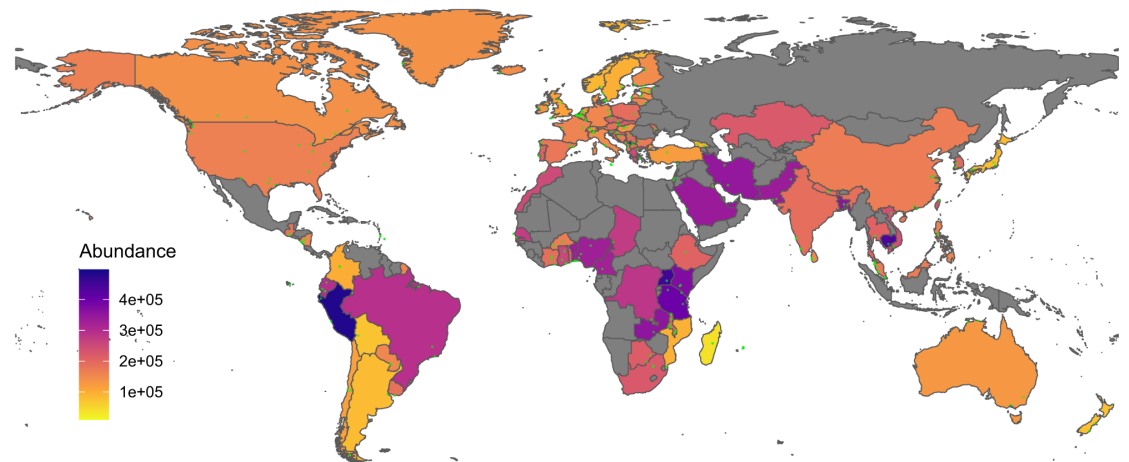
Patrick Munk , Christian Brinch, Frederik Duus Møller, Thomas N. Petersen, Rene S. Hendriksen, Anne Mette Seyfarth, Jette S. Kjeldgaard, Christina Aaby Svendsen, Bram van Bunnik, Fanny Berglund, Global Sewage Surveillance Consortium, D. G. Joakim Larsson, Marion Koopmans, Mark Woolhouse & Frank M. Aarestrup

*Nature Communications* 13, Article number: 7251 (2022) | [Cite this article](#)

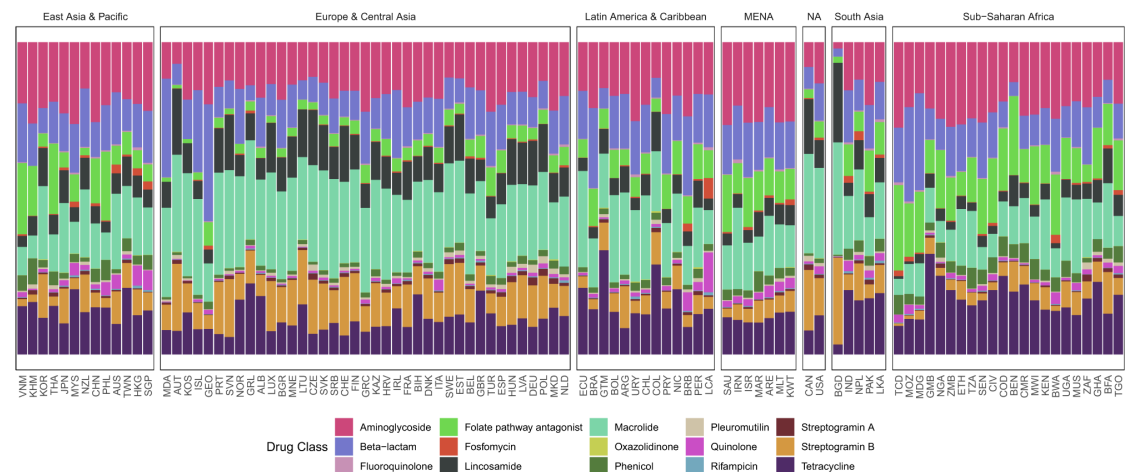
<https://doi.org/10.1038/s41467-022-34312-7>

- The global resistome based on sewage-based monitoring
- Metagenomic samples revealed regional patterns reflecting taxonomical changes
- Understanding the emergence, evolution, and transmission of individual antibiotic resistance genes (ARGs) is essential to develop sustainable strategies to fight this threat

a



b



# Microbial community at the Deepwater Horizon oil spill site

JOURNAL ARTICLE

## Metagenomics reveals sediment microbial community response to Deepwater Horizon oil spill

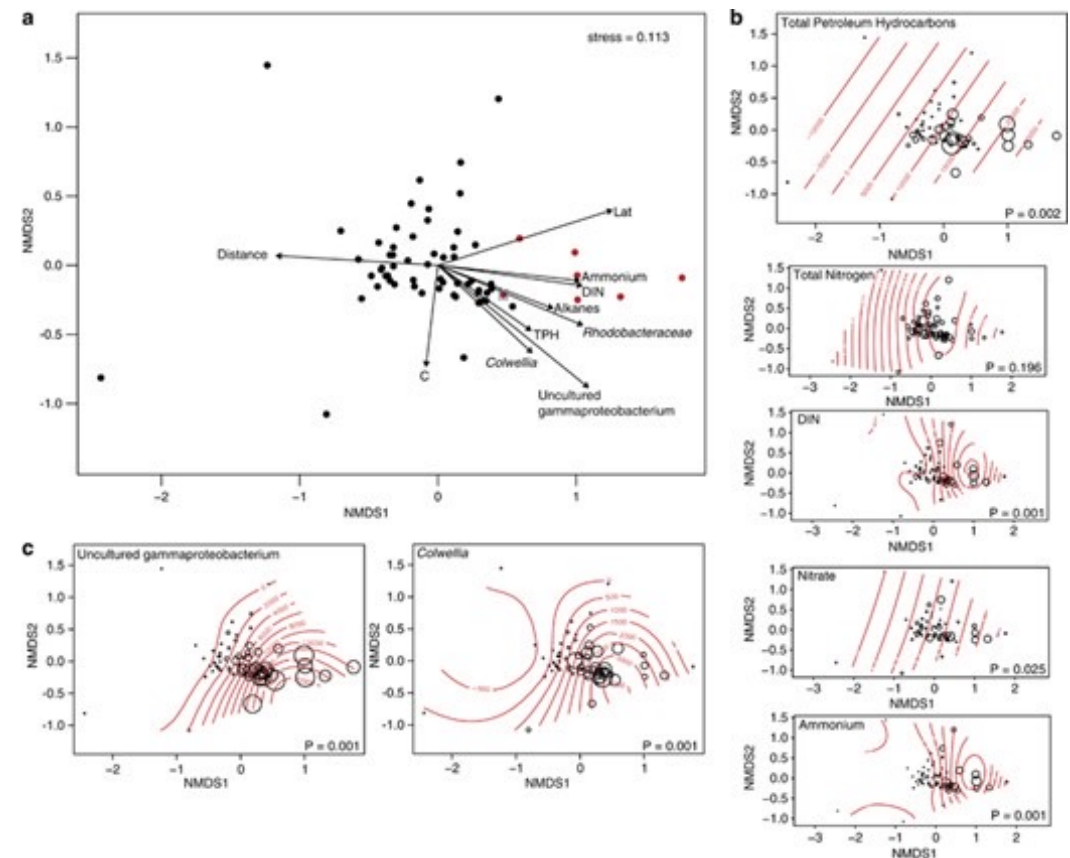


Olivia U Mason ✉, Nicole M Scott, Antonio Gonzalez, Adam Robbins-Pianka, Jacob Bælum, Jeffrey Kimbrel, Nicholas J Bouskill, Emmanuel Prestat, Sharon Borglin, Dominique C Joyner ... Show more

The ISME Journal, Volume 8, Issue 7, July 2014, Pages 1464–1475,

<https://doi.org/10.1038/ismej.2013.254>

- Metagenomic Analysis of sediment samples from 64 sites in the Gulf of Mexico, characterizing microbial community's response to the oil spill
- Uncultured *Gammaproteobacterium* and *Colwellia* species were enriched in oil-impacted sediments (hydrocarbon degradation)
- An increase in genes associated with hydrocarbon degradation pathways (aliphatic and aromatic hydrocarbon breakdown), indicating adaptive microbial response to oil contamination
- The study emphasized the importance of microbial communities in natural bioremediation processes.



# Global ocean microbiome (Tara oceans)

Article | [Open access](#) | Published: 11 June 2018

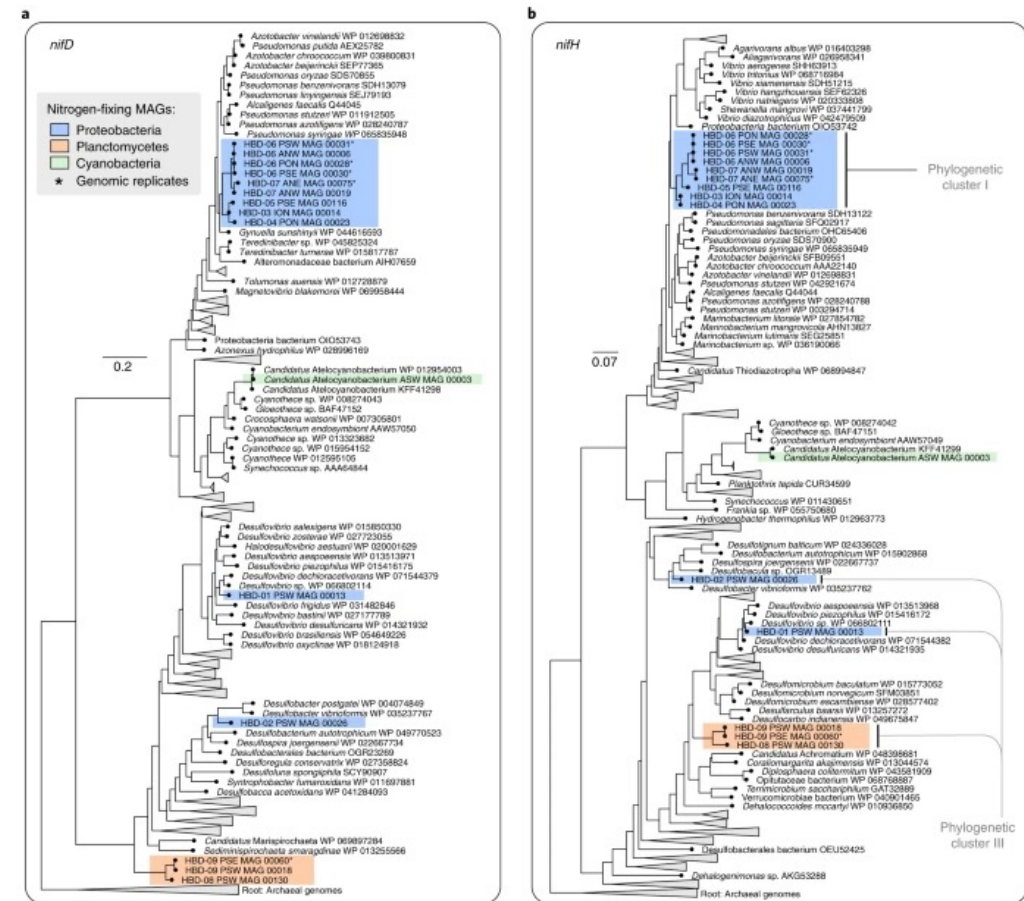
## Nitrogen-fixing populations of Planctomycetes and Proteobacteria are abundant in surface ocean metagenomes

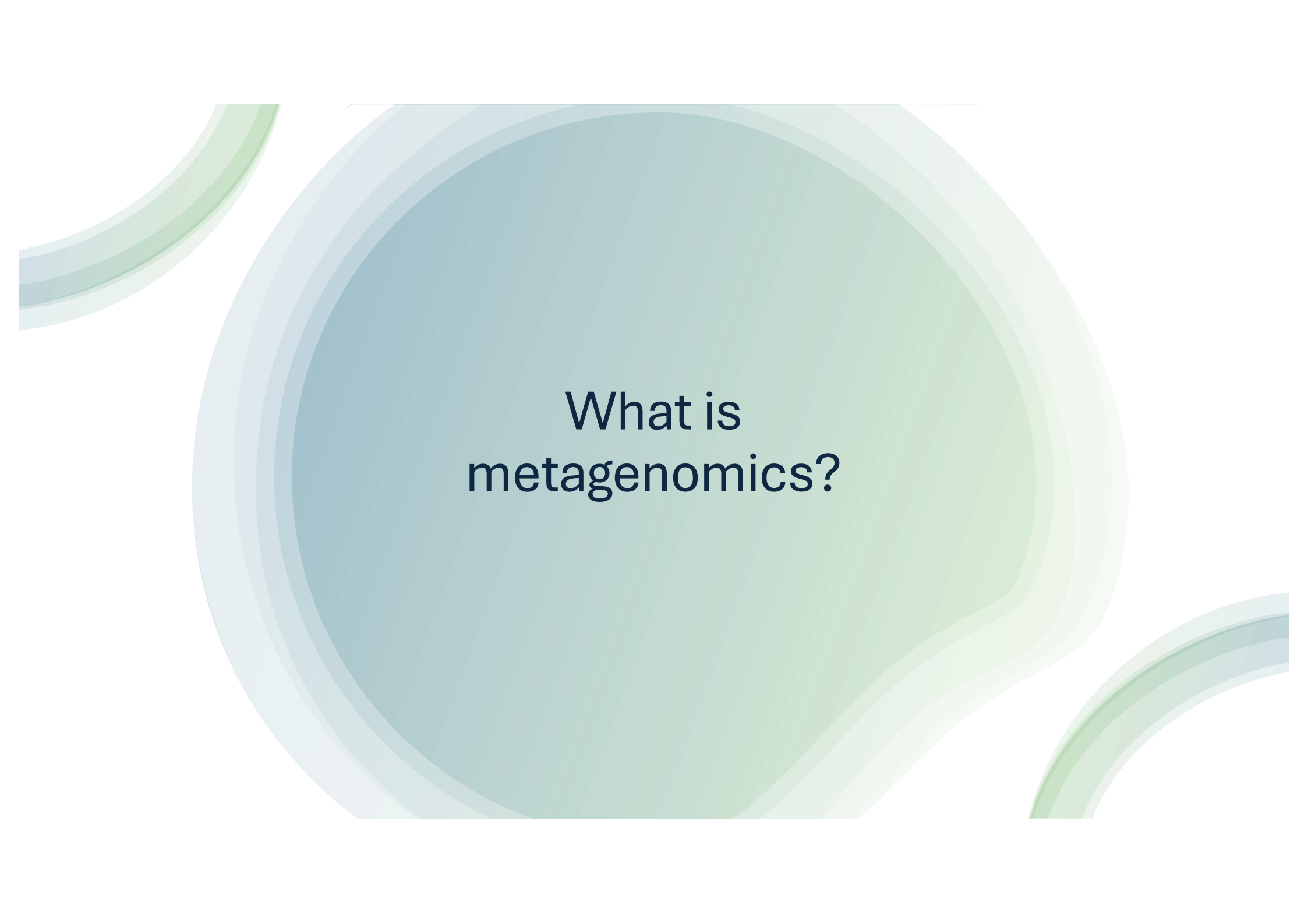
[Tom O. Delmont](#) , [Christopher Quince](#), [Alon Shaiber](#), [Özcan C. Esen](#), [Sonny TM Lee](#), [Michael S. Rappé](#), [Sandra L. McLellan](#), [Sebastian Lückner](#) & [A. Murat Eren](#) 

*Nature Microbiology* 3, 804–813 (2018) | [Cite this article](#)

<https://doi.org/10.1038/s41564-018-0176-9>

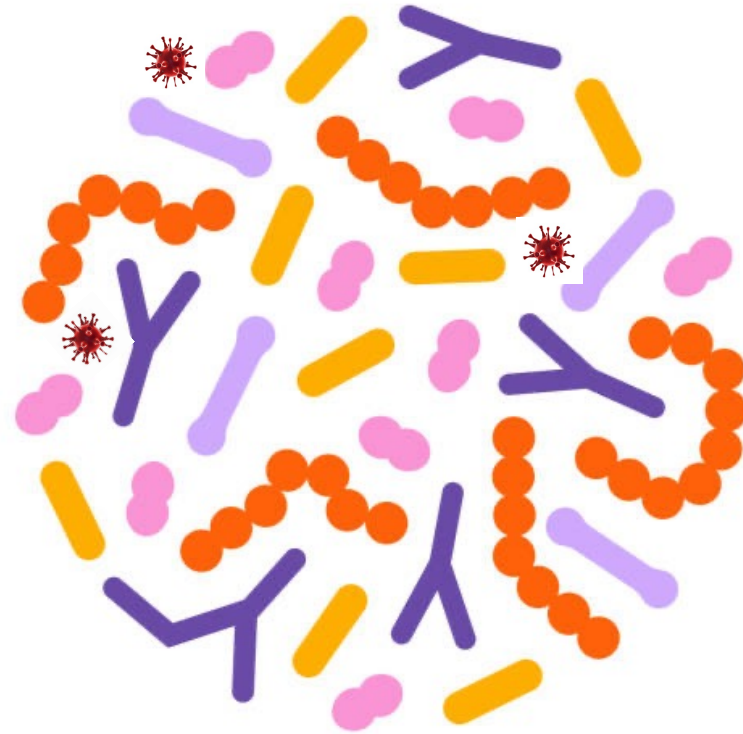
- Characterized ~1,000 non-redundant microbial genomes from the TARA Oceans metagenomes
- Nitrogen fixation in the surface ocean impacts global marine nitrogen bioavailability and thus microbial primary productivity
- Non-cyanobacterial diazotrophs inhabiting surface waters of the open ocean, which correspond to lineages within the Proteobacteria and Planctomycetes





What is  
metagenomics?

# Microbiome definition



# Microbiome definition

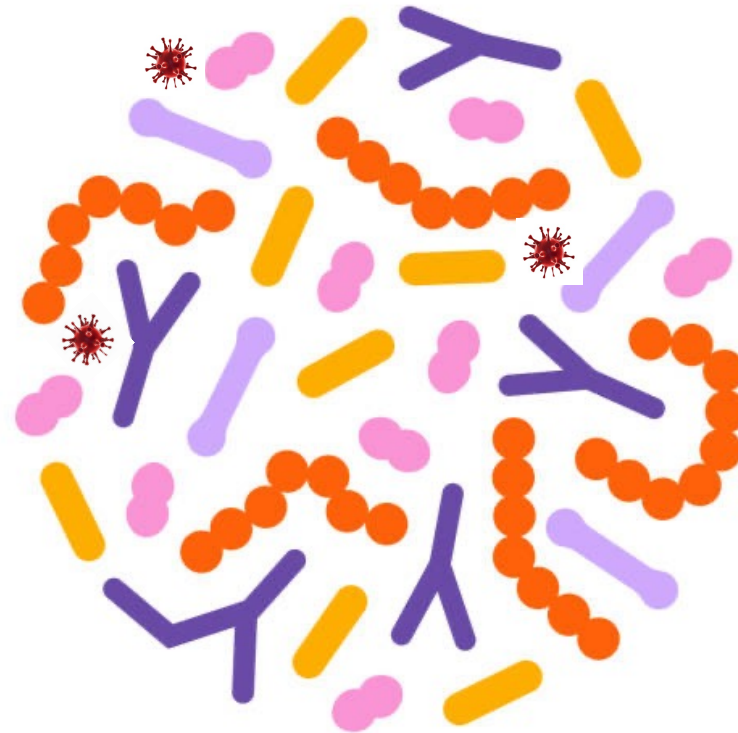
Microbiome refers to

Inhabitants:

- Bacteria
- Archaea
- Fungi
- Viruses
- Protozoa
- ...

Environment:

- Soil
- Ocean
- Wastewater plant
- Gut
- Skin
- ...



Organism products:

- Genes
- Proteins
- Pathways
- Metabolites

In contrast to a monoculture (all cells belong to one organism), in a microbiome we have cells of many different organisms → microbial community

What a microbiome study usually answer?

# What a microbiome study usually answer

Who is in the samples?

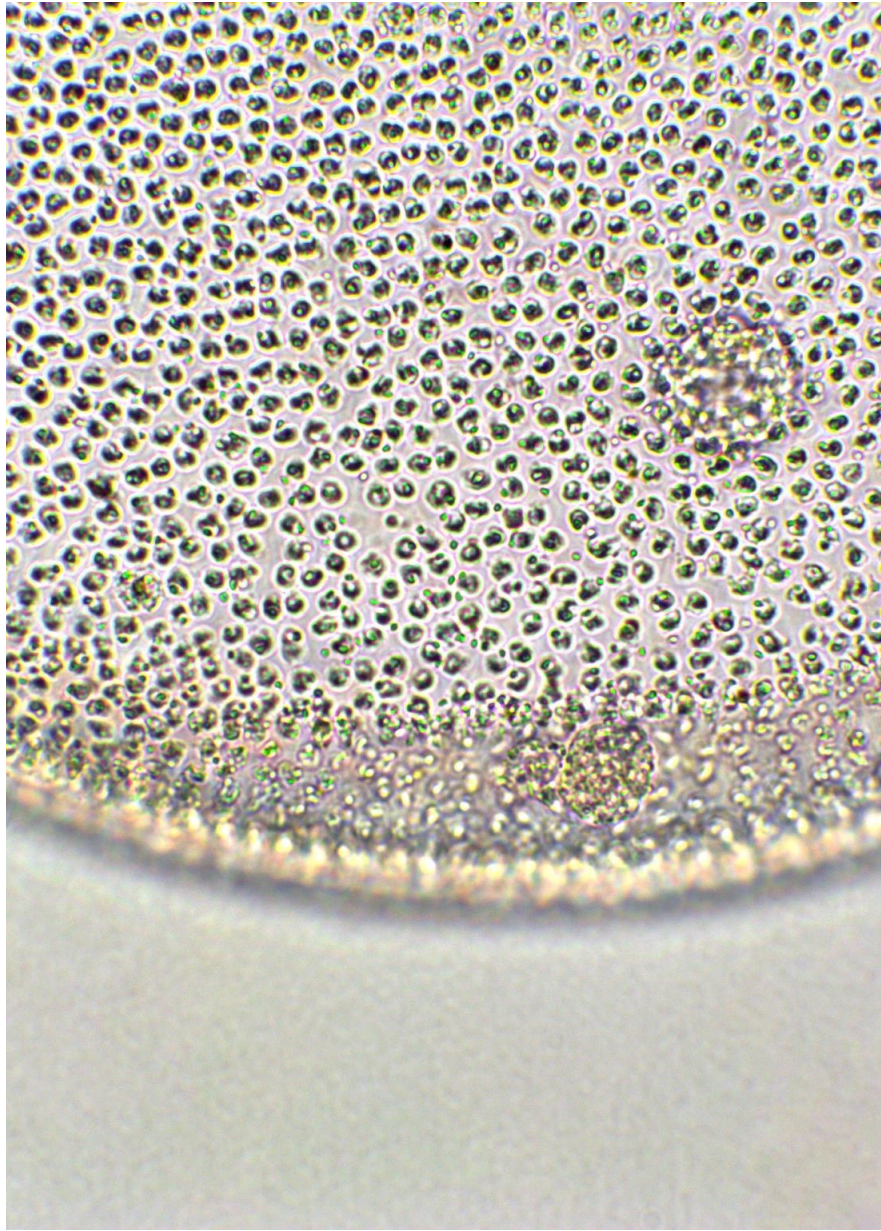
What is the relative proportion of the different members  
What is the total abundance (number of cells) (by absolute abundance)  
How abundant are bacteria, archaea, protozoa, virus,...  
How diverse is the microbial community  
How evenly distributed is it?

What are these organisms doing?

What functions potentially do they perform  
What substrates they consume?  
What metabolites they produce?

How do they perform their activities?

Which genes and/or pathways are present?

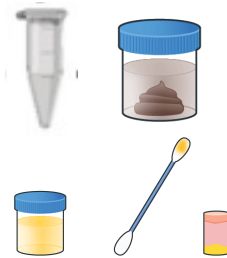
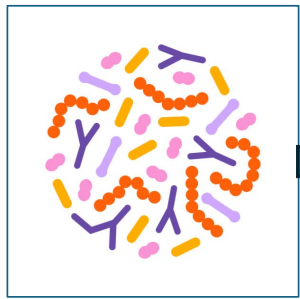


# What is Metagenomics?

- Study of the genetic material sampled directly from the environment (fecal, skin, soil, marine,... samples)
- No need of culturing organisms – many of the organisms that we characterize can not even be cultured
- Uses high-throughput sequencing to analyze microbial diversity, function, and microbial interactions

# Experimental design

**DNA Sampling**  
spatula, swabs,  
filtration units,  
soil cores,...



**DNA Storage**  
freezing,  
preservation  
buffers,...



**DNA Extraction**  
different kits



**DNA Sequencing**  
library prep - fragmented  
DNA tagged with adapters



Taxonomical and  
functional  
characterization of the  
sequenced reads  
(fastQ files)



Microbiome abundances

| OTU   | Sample 1 | Sample 2 | Sample 3 |
|-------|----------|----------|----------|
| A     | 0.60     | 0        | 0.70     |
| B     | 0.24     | 0.20     | 0.10     |
| C     | 0.10     | 0        | 0        |
| D     | 0.05     | 0        | 0        |
| E     | 0.01     | 0        | 0        |
| F     | 0        | 0.80     | 0.20     |
| Total | 1.0      | 1.0      | 1.0      |

# Sample collection

- To obtain a representative and uncontaminated snapshot of a microbial community in a specific environment (e.g., gut, soil, water, skin,...)
- Sterile tools and containers need to be used
- Microbial DNA can degrade rapidly. A good preservation ensures the microbial community structure remains as close to in situ as possible.
  - Freezing (-20 to -80C) for long-term storage
  - Preservation buffers if freezing is delayed
- Process quickly after sampling to reduce DNA degradation and community shifts if freezing is not possible
- For sample transportation use cold chain logistics (ice packs, dry ice, liquid nitrogen)
- Label clearly with metadata: time, location, environmental conditions, etc.
  - Important for downstream analysis and reproducibility



# DNA Extraction

- To obtain **high-quality, unbiased, and inhibitor-free DNA** from a complex matrix (e.g., soil, feces, water)
- Isolate **total microbial DNA** from all organisms in a sample (bacteria, archaea, fungi, viruses—depending on context)
- Steps:
  - **Cell Lysis:** The goal is to **lyse all types** of microbes, including tough ones like Gram-positive bacteria or spores.
    - Mechanical disruption (Bead beating)
    - Chemical/enzymatic lysis (detergent/enzymes (lysozyme, proteinase K, Rnase))

Kits are preferred in metagenomics for consistency, ease, and reduced contamination risk (e.g., Qiagen, Zymo, MoBio/DNeasy)

- **Contaminant Removal:** Use specialized kits with inhibitor removal steps (e.g., PowerSoil DNA Kit)
- **DNA Purification:** remove proteins, RNA, lipids, and inhibitors
- **Quantification & Quality Check**

Common DNA extraction kits:

- QIAGEN DNeasy PowerSoil Pro Kit
- QIAGEN QIAamp PowerFecal Pro DNA Kit
- ZymoBIOMICS DNA Miniprep Kit



# Sequencing

- To read the **nucleotide sequences** of DNA fragments extracted from a microbial community — enabling identification of **who is present** and **what they can do**
- **Library preparation:** DNA must be **fragmented** and **tagged with adapters** for compatibility with sequencing platforms.
- Include negative and positive **controls** (e.g., mock communities)
- Choose **depth** wisely: deeper for complex samples or analysis (e.g., soil, strain resolution, single nucleotide variants(SNVs)), shallower for simpler ones
- <https://www.youtube.com/watch?v=HMyCqWhwB8E>



# Platforms

| Platform               | Features                   | Read Length   | Use Case                           |
|------------------------|----------------------------|---------------|------------------------------------|
| <b>Illumina</b>        | High accuracy, short reads | ~150–300 bp   | Most common for shotgun & amplicon |
| <b>Oxford Nanopore</b> | Long reads, portable       | up to 100 kb+ | Assembly, rare taxa detection      |
| <b>PacBio HiFi</b>     | Long + accurate reads      | ~10–25 kb     | Functional profiling, assembly     |
| <b>Ion Torrent</b>     | Medium throughput          | ~200–400 bp   | Less common but used in clinical   |



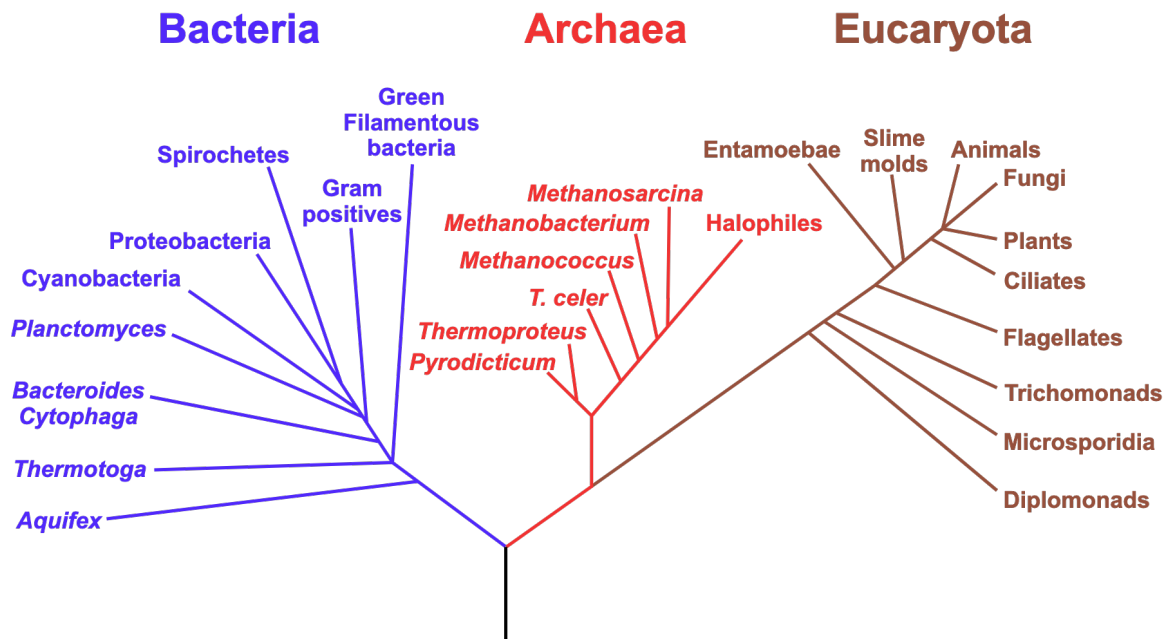
# Taxonomical profiling

# Taxonomical profiling

- Process of identifying and quantifying the microbial taxa present in a sample - this can be bacteria, archaea, viruses, fungi, ...
- Typically from metagenomic or amplicon sequencing data (metabarcoding).
- The most common taxonomical profiling methods can be grouped based on the type of sequencing data used:
  - Amplicon based profiling: 16S, 18S, ITS rRNA sequencing – e.g. 16S rRNA gene is used to profile bacteria and archaea
  - Shotgun Metagenomics-based Profiling: whole-genome sequencing data
    - Read-based methods: direct mapping of reads to genes, proteins or genomes
    - Assembly-based methods

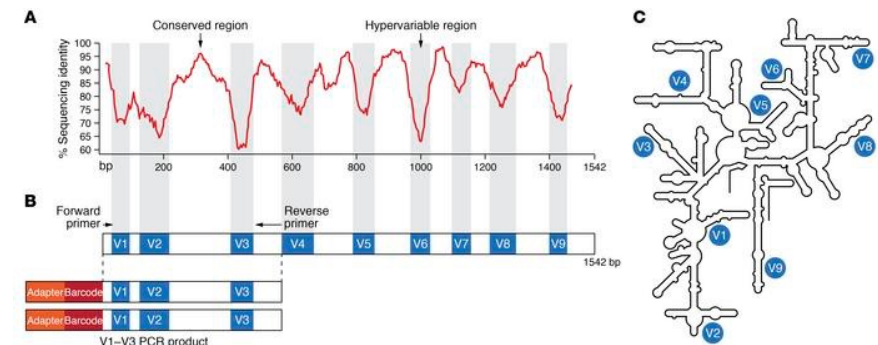
# 16S amplicon sequencing

## Phylogenetic Tree of Life



Universal phylogenetic tree based on the 16S rRNA gene sequence comparisons.  
 Pace, N. A molecular view of microbial diversity and the biosphere.  
 Science 276:734-740. 1997.

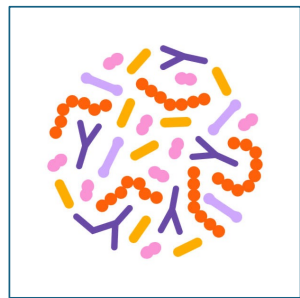
- 16S rRNA gene is an ancient gene that encodes the RNA component of the small (30S) subunit of the prokaryotic ribosome
- present also in mitochondria and chloroplasts of eukaryotes
- traditionally used to infer phylogenetic trees to figure out species evolution
- Conserved and variable regions (easy to find and amplify with some unique variable regions to tag species)



*J Clin Invest.* 2022;[132\(7\)](https://doi.org/10.1172/JCI154944):e154944.  
<https://doi.org/10.1172/JCI154944>.

# 16S amplicon sequencing

- Targets specifically 16S rRNA
- PCR amplification



16S rRNA  
gene

amplicon  
sequencing

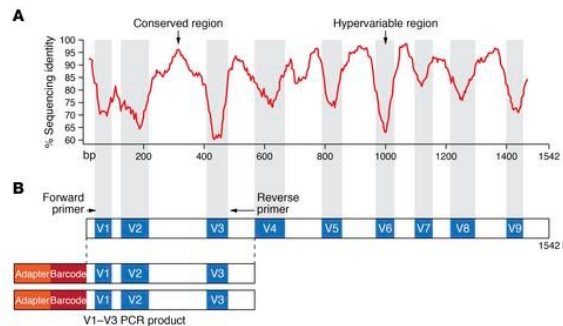
CGCCTAT  
CACCTAT  
CGCCTAT  
CGCCTAC  
CGCCTAT

Compare to a  
reference database



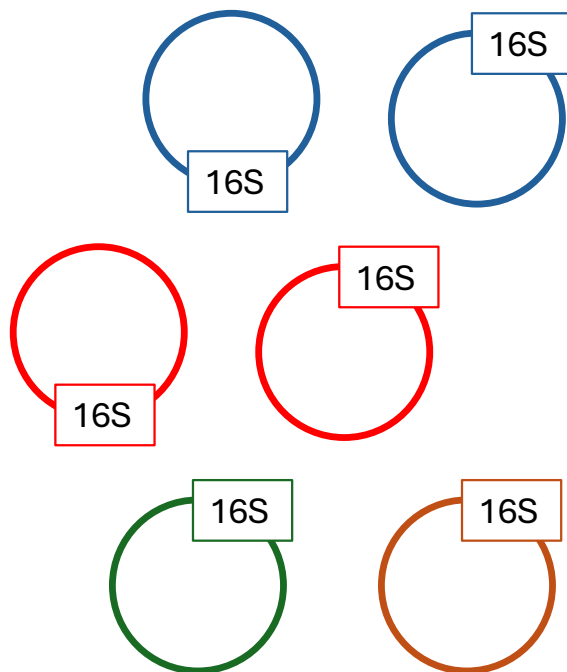
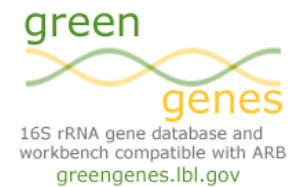
Abundance table

| Features | Samples |  |  |  |  |  |  |  |  |  |
|----------|---------|--|--|--|--|--|--|--|--|--|
|          |         |  |  |  |  |  |  |  |  |  |
|          |         |  |  |  |  |  |  |  |  |  |
|          |         |  |  |  |  |  |  |  |  |  |
|          |         |  |  |  |  |  |  |  |  |  |

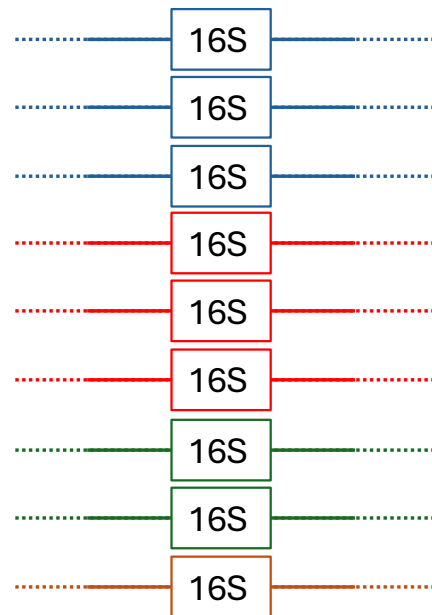


- Operational taxonomic Units (OTUs) – Resolution up to Genus
- Inferred functions

# 16S amplicon sequencing



Microbial genomes in our sample carrying 16S rRNA gene



PCR amplification of 16S rRNA gene to produce a pool of DNA containing only 16S genes

```

TTCAACCGAGATATGC
TTCAACCGAGATATGC
TTCAACCGAGATATGC
TTTAACCGATATATGC
TTTAACCGATATATGC
TTTAACCGATATATGC
TTTAACAGAGATATGC
TTTAACAGAGATATGC
TTTAACCGAGATATAC
    
```

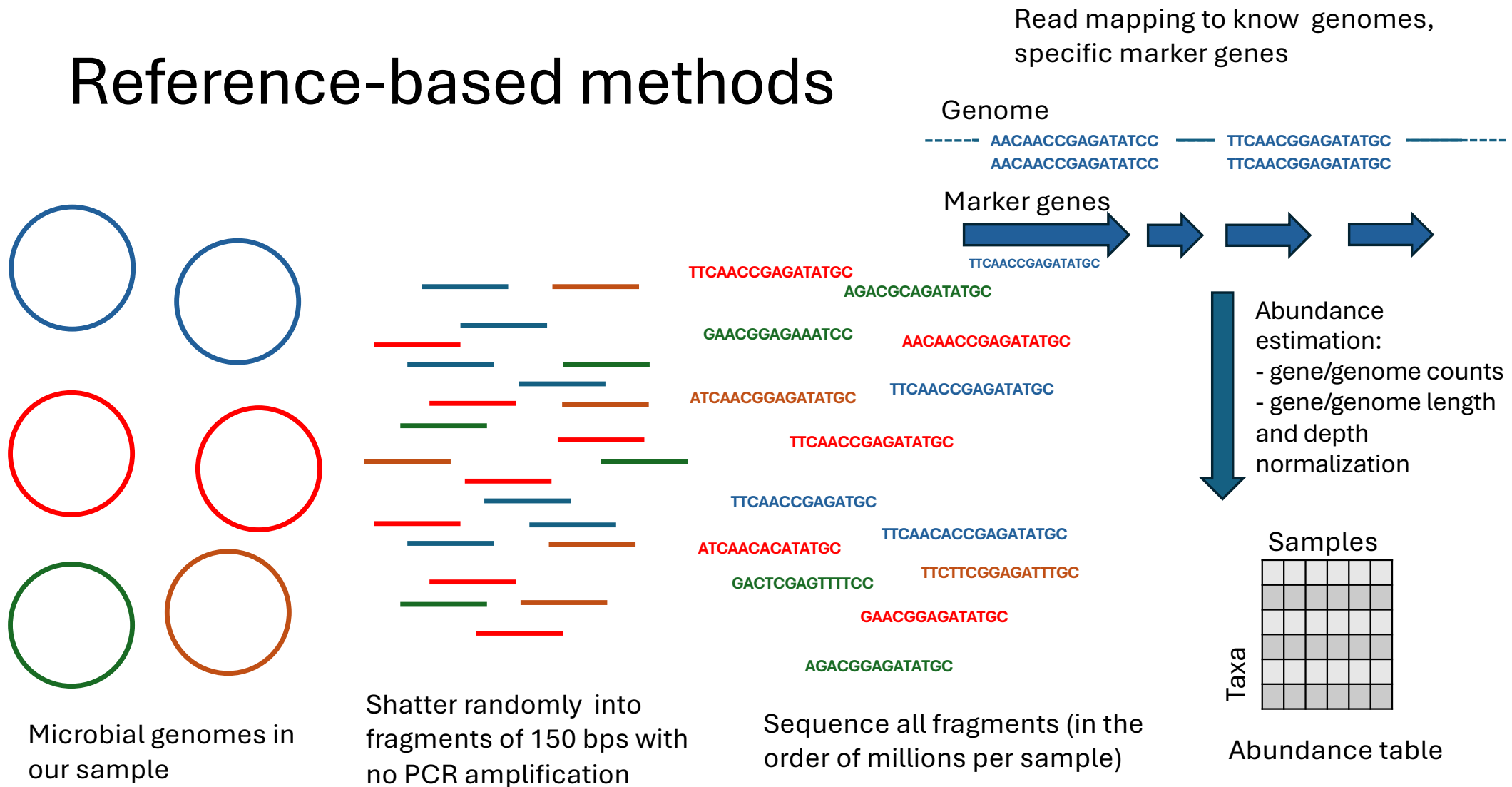
Sequencing all fragments

Bin by operational taxonomic units and compare to known 16S databases



Abundance table of operational taxonomic units (OTUs) – resolution up to Genus

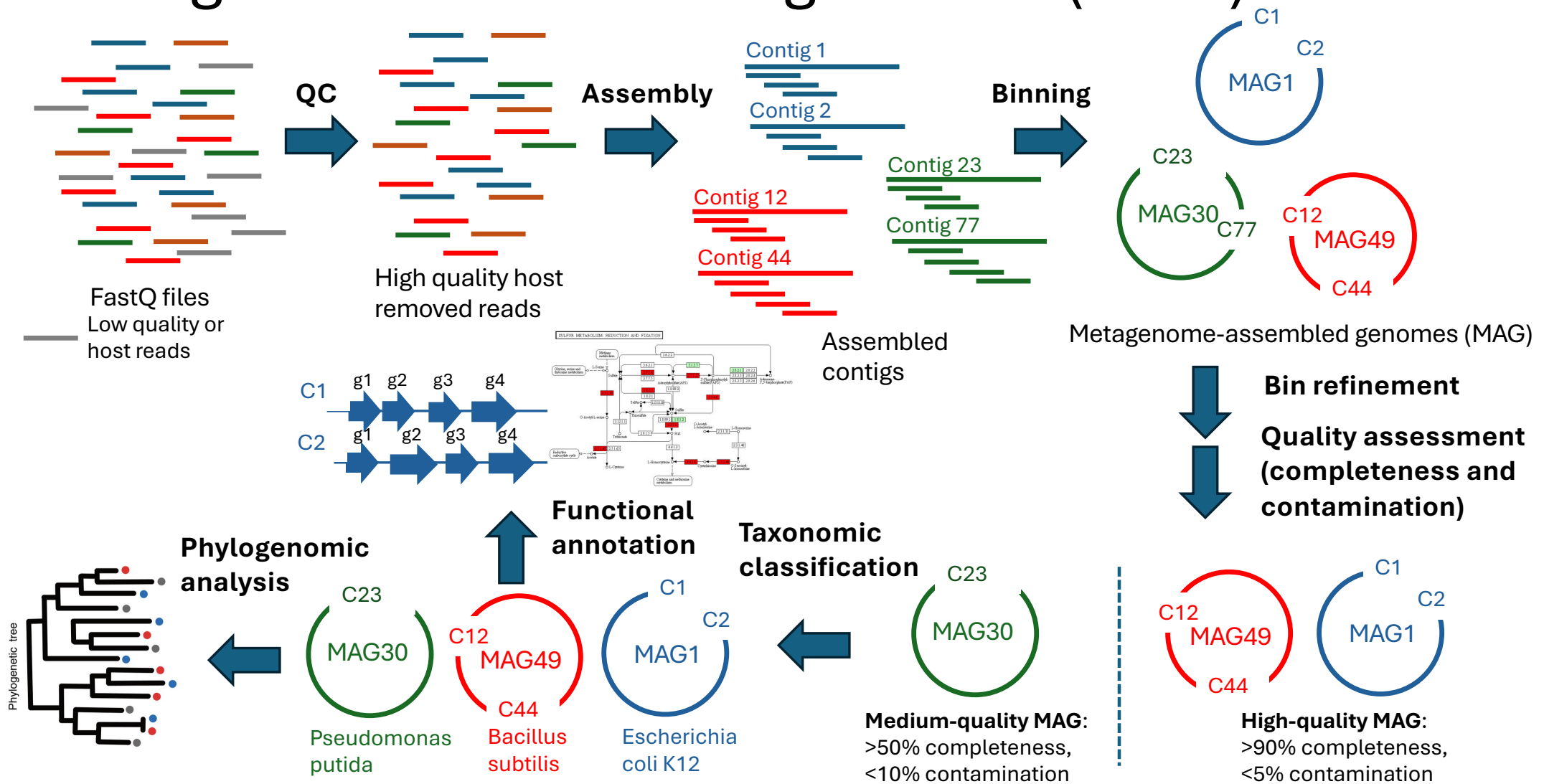
# Reference-based methods



# Amplicon and read-based shotgun metagenomics tools

| Method                                                           | Tools                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Databases                                                                                                                                                     |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Amplicon-based Profiling                                         | <p><b>QIIME 2:</b> A popular pipeline supporting DADA2, Deblur, and multiple classifiers (Naive Bayes, etc.).</p> <p><b>DADA2:</b> High-resolution method that infers amplicon sequence variants (ASVs).</p> <p><b>USEARCH / VSEARCH:</b> Clustering-based OTU picking methods.</p> <p><b>Mothur:</b> OTU-based method using different clustering and alignment strategies.</p> <p><b>RDP Classifier:</b> Naive Bayesian classifier for assigning taxonomy.</p>                           | <p><b>SILVA</b></p> <p><b>Greengenes</b> (deprecated but still used)</p> <p><b>RDP</b></p> <p><b>UNITE</b> (for fungi)</p>                                    |
| Shotgun Metagenomics-based Read-based (Direct Mapping) profiling | <p><b>MetaPhlAn4:</b> Uses clade-specific marker genes for fast, accurate profiling.</p> <p><b>Kraken2 / KrakenUniq:</b> K-mer based, ultra-fast classification against a database of genomes.</p> <p><b>Centrifuge:</b> K-mer based, designed for large reference databases and low-memory use.</p> <p><b>HUMAnN / HUMAnN3:</b> Often used alongside MetaPhlAn for functional and taxonomic profiling.</p> <p><b>StrainPhlAn:</b> Extension of MetaPhlAn for strain-level profiling.</p> | <p><b>ChocoPhlAn</b> (used by MetaPhlAn)</p> <p><b>RefSeq / GenBank</b> (used by Kraken/Centrifuge)</p> <p><b>GTDB</b> (Genome-based taxonomic reference)</p> |

# Metagenome-assembled genomes (MAG)



# MAG construction tools

| Step                     | Goal                                                                   | Tools                                     | Notes                                                                                                                                             |
|--------------------------|------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality Control          | Remove low-quality reads, adapters, and contaminants (e.g. host reads) | fastp, adapterremoval, Trimmomatic, BBduk | Output: Clean reads                                                                                                                               |
| Assembly                 | Assemble short reads into longer contigs.                              | MEGAHIT, metaSPAdes                       | Output: Assembled contigs                                                                                                                         |
| Binning                  | Group contigs that belong to the same genome                           | MetaBAT2, MaxBin2, CONCOCT, VAMB          | Output: bins of contigs belonging to the same genome<br>Using coverage profiles and sequence composition (GC content, tetranucleotide frequency). |
| Bin refinement           | Improve bin quality and merge results from different binning tools     | DASTool, metaWRAP                         | Output: Refined bins                                                                                                                              |
| Quality Assessment       | Assess completeness and contamination of bins (MAGs)                   | CheckM, BUSCO                             | Metrics usually used:<br>High-quality MAG: >90% completeness, <5% contamination.<br>Medium-quality MAG: >50% completeness, <10% contamination.    |
| Taxonomic classification | Assign taxonomy to MAGs                                                | GTDB-Tk, Kraken2, CAT/BAT                 | Output: taxonomically annotated MAGs                                                                                                              |
| Functional annotation    | Predict genes and metabolic functions                                  | Prokka, EggNOG-mapper                     | Out put: functionally annotated MAGs                                                                                                              |
| Phylogenomic analysis    | Compare MAGs evolutionarily                                            | PhyloPhlAn, Anvi'o, IQ-TREE               | Output: Phylogenetic trees based on MAGs                                                                                                          |

# Summary table

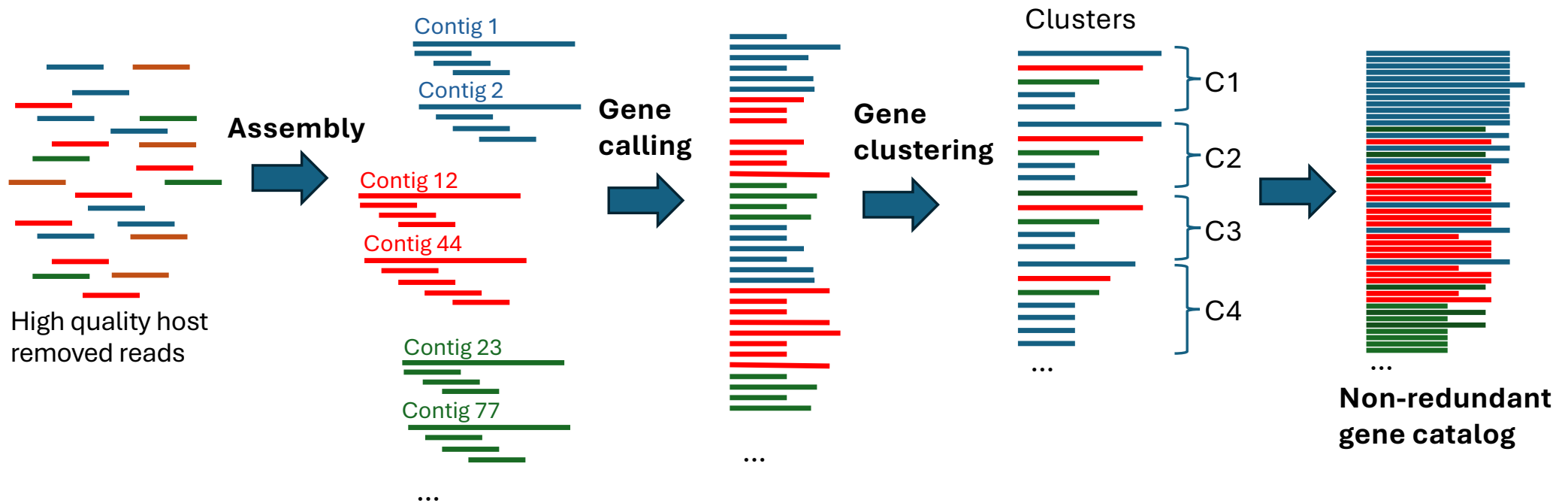
|      | Amplicon/16S sequencing                                                                                                                                                                                                                 | Shotgun metagenomics                                                                                                                                                                                                                       |                                                                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                                                                                                                                         | Mapping - based                                                                                                                                                                                                                            | Assembly - based                                                                                                                                                                                                                              |
| Pros | <ul style="list-style-type: none"> <li>• Can handle low biomass / non-bacterial contamination</li> <li>• Rough taxonomy of novel species</li> <li>• Less expensive</li> </ul>                                                           | <ul style="list-style-type: none"> <li>• High resolution (subspecies, genes SNVs)</li> <li>• Can detect all DNA in the sample (also virus, fungus, protozoa,...)</li> <li>• Fast and scales well</li> <li>• Finds rare entities</li> </ul> | <ul style="list-style-type: none"> <li>• High resolution (subspecies, genes SNVs)</li> <li>• Can detect all DNA in the sample (also virus, fungus, protozoa,...)</li> <li>• Can identify novel genomes, genes and synteny patterns</li> </ul> |
| Cons | <ul style="list-style-type: none"> <li>• Low resolution</li> <li>• Limited to bacteria or archaea</li> <li>• No subspecies or strains resolution</li> <li>• Biases due to PCR amplification / amplicon copy number variation</li> </ul> | <ul style="list-style-type: none"> <li>• Does not work well for low biomass</li> <li>• More expensive (currently very competitive prices)</li> <li>• Limited to what is known in the databases</li> </ul>                                  | <ul style="list-style-type: none"> <li>• Does not work well for low biomass</li> <li>• More expensive (currently very competitive prices)</li> <li>• Misses rare entities</li> <li>• Computationally very intensive</li> </ul>                |



# Functional profiling

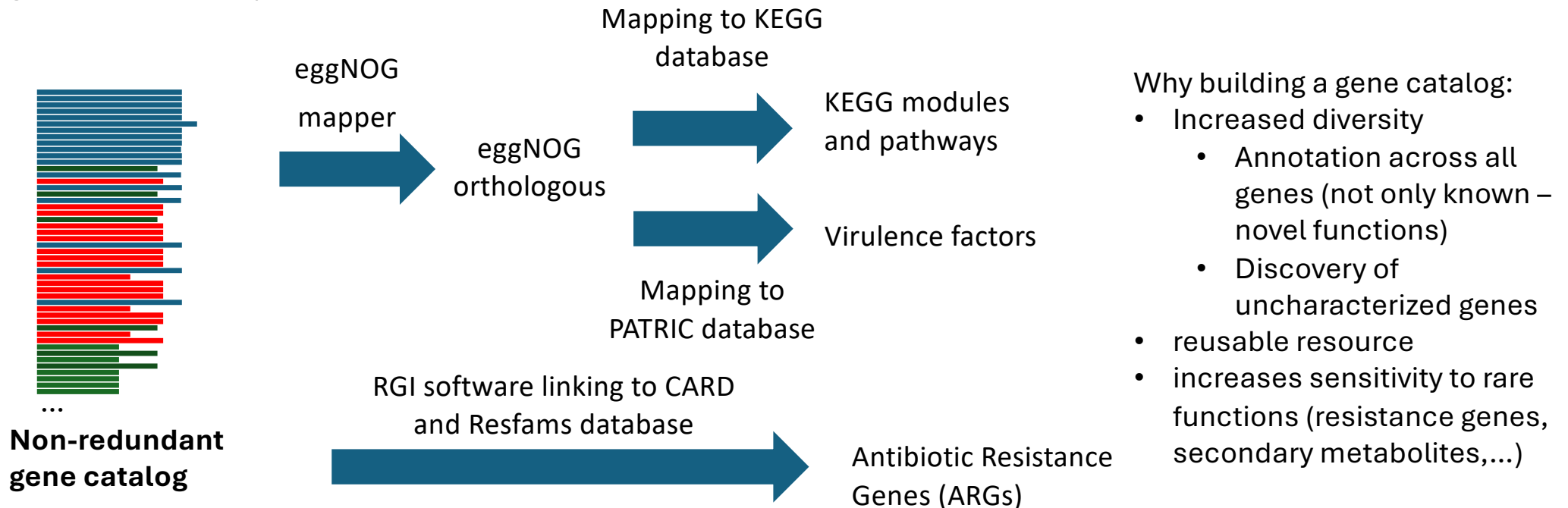
# Functional annotation from the gene catalog

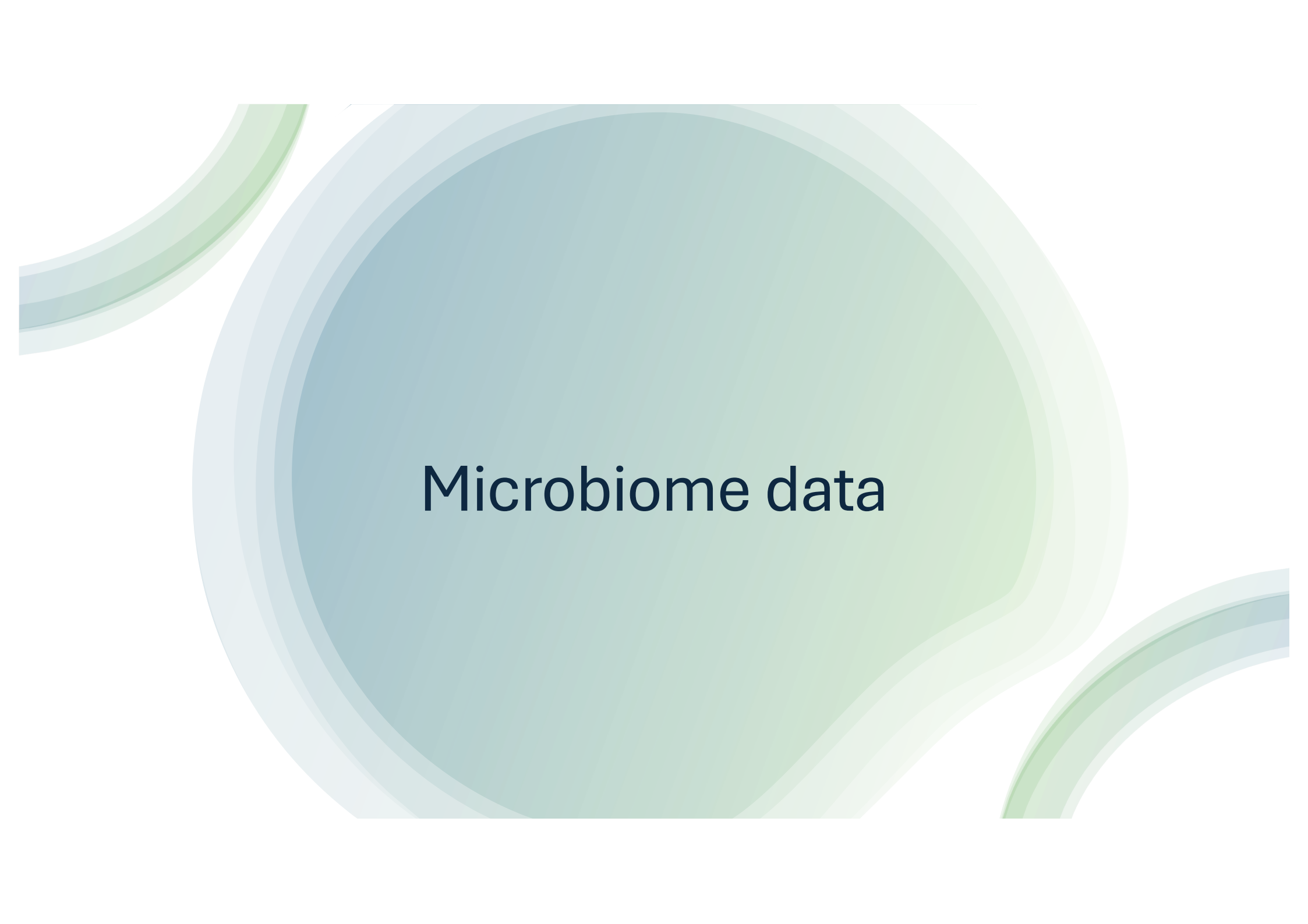
From both read-based and assembled-based shotgun metagenomics we end up on lists of genes that we can cluster and turn into a non-redundant gene catalog



# Functional annotation from the gene catalog

Reads can be mapped to the catalog genes to infer gene abundance and those genes can be mapped to several functional information databases (e.g KEGG database, PATRIC Database, CARD or Resfams antibiotic resistance gene databases,...)





# Microbiome data

# Microbiome data

- **Sparsity**
- **Dynamic range**
- Usually expressed as Relative, not Absolute abundance
- Compositional data

|                                                  |           | Relative abundance (%) |      |     |     |      |    |     |    |   |
|--------------------------------------------------|-----------|------------------------|------|-----|-----|------|----|-----|----|---|
| Area                                             | Condition | 1                      | 2    | 3   | 1   | 2    | 3  | 1   | 2  | 3 |
|                                                  |           | X                      | X    | X   | Y   | Y    | Y  | Z   | Z  | Z |
| Relative abundance<br>Taxa or functional feature | 10        | 11                     | 20   | 12  | 14  | 16   | 25 | 34  | 22 |   |
|                                                  | 70        | 49                     | 56   | 0.9 | 0.5 | 0    | 0  | 0.1 | 0  |   |
|                                                  | 0         | 0                      | 0    | 0.4 | 0.1 | 0    | 0  | 0   | 0  |   |
|                                                  | 0.3       | 0.4                    | 0.45 | 0   | 0.1 | 1    | 2  | 3   | 4  |   |
|                                                  | 2         | 2                      | 3    | 0   | 0   | 0.01 | 1  | 1   | 2  |   |
|                                                  |           |                        |      |     |     |      |    |     |    |   |
|                                                  |           |                        |      |     |     |      |    |     |    |   |
|                                                  |           | Samples                |      |     |     |      |    |     |    |   |

# Microbiome data

Microbiome data is compositional:

- Sparsity
  - Dinamic range
  - **Relative vs Absolute abundance**
  - **Compositional data**
- Sampling of a fraction of the total genomic DNA
  - Different sequencing depth between samples
  - Both sampling depth and sequencing depth can vary several orders of magnitude between samples
  - Absolute abundances are normally not known – abundances are relative and constraint to sum to 1 (or 100%)
  - Relative abundances are susceptible to compositional effects

# Compositional data

Consider read counts as a proxy for cell abundance (not perfect as in 16S we have multiple 16S gene copies, and in shotgun metagenomics we have different genome sizes – larger genomes contributing with more reads per cell), extraction efficiency, library prep bias ..)

|        | S1 cells | S1 Rel ab % | S2 cells | S2 Rel ab % |
|--------|----------|-------------|----------|-------------|
| Taxa 1 | 250      | 25          | 250      | 33.33       |
| Taxa 2 | 250      | 25          | 250      | 33.33       |
| Taxa 3 | 300      | 30          | 50       | 6.67        |
| Taxa 4 | 200      | 20          | 200      | 26.67       |
| Total  | 1000     | 100         | 750      | 100         |

- Only Taxon 3 reduced its abundance after antibiotic treatment, but 1,2, and 4 seem that they increase
- Affects downstream analysis and interpretations
  - e.g. correlations or regression assume independent variables
  - other analysis alternatives have been proposed, center-log-ratio transformation (clr), ALDEx2, ANCOM
  - however, they introduce practical limitations — especially with zero handling, computational cost, and difficult interpretation

# Ideal scenario: absolute abundances

If we want to know **true cell counts**, we need to:

- Add **internal spike-in controls** (known amounts of reference DNA).
- Use **qPCR or flow cytometry** to estimate **total bacterial load**.
- This allows you to correct read counts to reflect **absolute abundance**, not just proportions.



# Diversity measures

# Diversity measures:

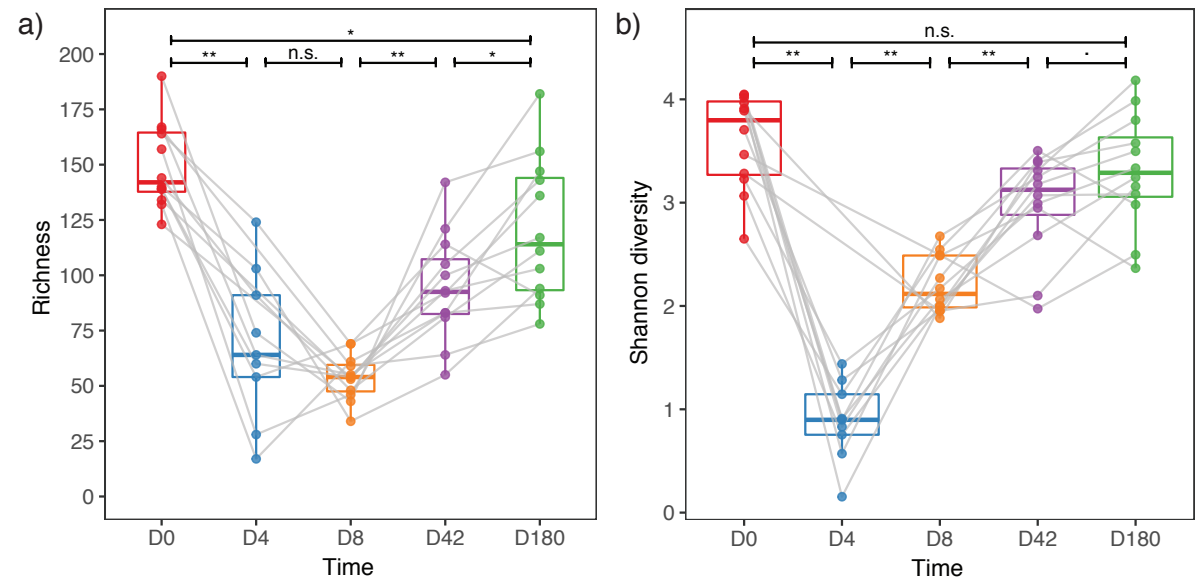
## Alpha diversity

- Shannon index:

$$H' = - \sum_{i=1}^S p_i \cdot \ln(p_i)$$

- S = total number of species
- $p_i$  = proportion of species  $i$  in the sample

- Diversity within a sample
  - Richness:** Number of unique taxa (e.g., species, OTUs, ASVs)
    - How many different taxa are in a sample
  - Shannon Index:** Accounts for both richness and evenness
    - Interprets:** Both richness and evenness (how balanced the community is)
    - Higher value** = More diverse and evenly distributed community
  - Other indices: Simpson, Chao1, Faith's PD (for phylogenetic diversity)

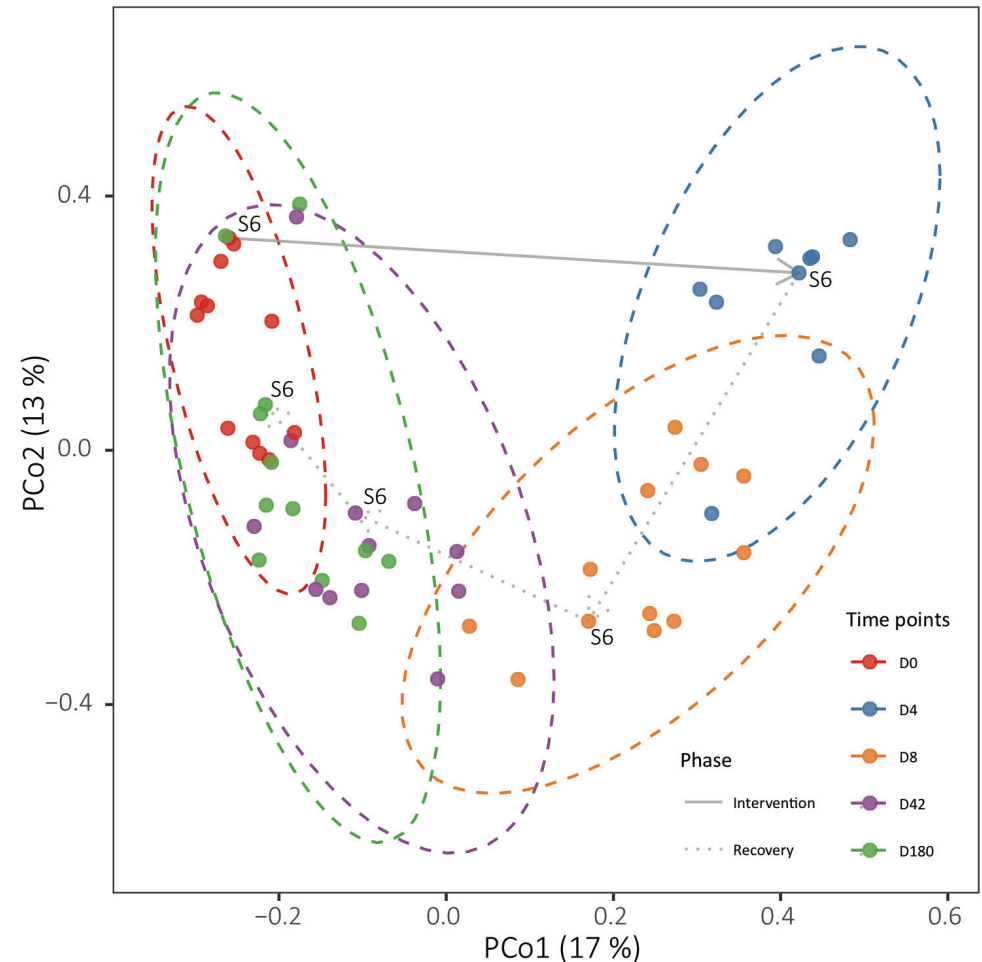


# Diversity measures:

## Beta diversity

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- Diversity between samples (community composition dissimilarity)
- For example:
  - **Bray-Curtis dissimilarity** (abundance-weighted)
  - **Jaccard distance** (presence/absence)
  - **UniFrac** (phylogenetic distances: weighted/unweighted)
- These distances between samples are calculated based on the taxa abundance table and projected on the two first dimensions of a PCoA. (other ordination methods are also used: NMDS, tSNE or simply PCA,...)



The background features a gradient from light blue on the left to light green on the right. Overlaid on this are several concentric, wavy, semi-transparent shapes in shades of blue and green, creating a layered, organic effect.

# Challenges and limitations

# Challenges and limitations of metagenomics studies

- Shotgun metagenomics have higher cost than 16S amplicon but you get higher taxonomical resolution and functional information
- Sampling biases can occur when collecting environmental samples which can also lead to misrepresentation of the microbial community
- It is also challenging to extract microbes DNA from low biomass samples - highly dependant on the sampling method and the its accuracy
- Microbial genomes available and deposited in reference databases are biased towards model organisms, pathogens and cultivable bacteria
- Metagenomic sequencing can detect the presence of microorganisms but cannot easily determine if they are pathogenic (do they have their toxins? Are they active?)
- Functional profiling is hindered by the lack of annotation for most genes
- “Microbial dark matter” – microbes not yet characterized can be recovered by assemble-based methods, but still many reads are not used in the assembly – sequencing noise, DNA contamination, and bacteria and plasmids that remain obscure even after some parts of their genomes are assembled
- “Live or dead” dilemma – DNA persists in the environment after the death of the host cell, sequencing results may not be representative of the active microbial population
- Snapshot of the microbiome - many metagenomics studies are cross-sectional and can not assess intra- vs inter-subject variation – more longitudinal studies are needed
- Disentangling the cause from the effect – follow ups, longitudinal and prospective studies needed
- Metagenomics have not yet reached the level of standardization of other high-throughput techniques